Modeling Terrain-Vehicle Interaction for Off-Road Mobility Assessment

***Abstract*—This work presents an integrated framework for forecasting terrain trafficability and planning safe routes for military ground vehicles. A hybrid GRU–Ridge model predicts soil strength using historical weather and soil-moisture data over a 1–7-day horizon. The predictions are combined with vehicle-specific mobility requirements and terrain characteristics to classify terrain and plan routes that avoid impassable ground, accounting for convoy-induced soil degradation. Experiments on the Pachmarhi plateau achieve an R^2 of 0.922 for soil strength and, compared with current-condition planning, reduce trips encountering impassable terrain by 27% and trips held back unnecessarily by 22%, all at seven days. The framework supports predictive, risk-aware military mobility planning under changing environmental conditions.**

***Index Terms*—Terrain trafficability, Soil strength forecasting, GRU–Ridge model, Vehicle Cone Index**

I. Introduction

Military ground mobility depends on vehicle characteristics and changing terrain conditions. Rainfall and soil-moisture variations can reduce soil strength, rendering planned routes impassable. However, existing trafficability methods largely assess current conditions rather than predict future terrain accessibility [1], [2].

The Rating Cone Index (RCI) measures soil bearing strength, while the Vehicle Cone Index (VCI) represents vehicle-specific soil strength requirements. Although these indices underpin established mobility models [3], [4], integrating soil-strength forecasting with vehicle-specific trafficability assessment and route optimization remains a challenge.

This study proposes an integrated framework combining a hybrid Gated Recurrent Unit–Ridge model for 1–7-day RCI forecasting with vehicle-specific VCI, terrain characteristics, and Dijkstra-based route planning. The main contributions of this work are as follows:

- A forecast of soil strength for the day of travel, 1–7 days ahead, instead of planning on current conditions and assuming that they persist. The hybrid GRU–Ridge forecaster is lightweight, with a sixth of the parameters of the only tested model with a lower average error.
- The grouping of vehicles into five bands (A–E) by their one-pass VCI, each with its median requirements and its own safety margin against forecast error per lead time, so that terrain is classified as GO, SLOW-GO, or NO-GO per class at a fixed miss rate, with SLOW-GO quantified by the number of vehicles one track can carry and the speed the soil allows.
- Fuzzy trafficability maps for each band on the day of travel, which combine the forecast soil strength with slope, land cover, and roads, Dijkstra-based routing on these maps, and a wear-aware convoy planner in which each vehicle plans its route on the track left by those before it.

II. Related Works

Military terrain trafficability assessment commonly relies on the RCI and VCI, supported by established models such as the NATO Reference Mobility Model [3]–[5]. Recent studies have developed terrain trafficability maps using environmental factors and route-planning algorithms [2]. GIS-based multi-criteria decision analysis maps off-road trafficability or land capability from elevation, land cover and soil data [1], [6], [7], path-planning studies select routes over road networks under uncertain conditions [8], [9], and lightweight detectors classify military vehicles on edge devices [10] (Table I). Meanwhile, machine-learning methods, including recurrent neural networks and Transformers, have been applied to soil-moisture forecasting [11], [12].

Despite these advances, none of these approaches forecasts soil strength and links it directly to vehicle-specific requirements and route planning, including terrain degradation during convoy movement, which the present study addresses.

TABLE I
Comparison of existing approaches. Light-weight: a compact model (ours: 531K parameters) or one shown to run on edge hardware.

Reference	Comparative Factors			
	Dataset	Terrain-Features	Light-weight	Vehicle-specific mobility
[1]	Sentinel-2, ALOS PALSAR, soil maps	✓	✗	✗
[6]	DEM-derived elevation and slope	✓	✗	✗
[7]	Sentinel-2 land cover, SRTM DEM, OSM roads, soil surveys	✓	✗	✗
[8]	GPS and sensor traffic data	✗	✗	✗
[9]	IEEE CEC path-planning benchmarks	✗	✗	✗
[10]	Google Maps and ISRO Bhuvan imagery	✗	✓	✗
Ours	ERA5-Land, ERA5 and IMD weather, Earth Engine terrain, OSM roads, 1,145 vehicles	✓	✓	✓

III. Problem Statement

The problem addressed in this study is to predict future terrain trafficability and determine safe, efficient routes for heterogeneous ground vehicles under changing environmental conditions. This requires estimating soil bearing strength several days in advance, relating it to each vehicle's mobility requirements, and accounting for terrain slope, land cover, road accessibility, and degradation from repeated vehicle use.

A central challenge is balancing operational safety with mobility: overly optimistic assessments risk sending vehicles into impassable terrain, while overly conservative assessments unnecessarily restrict movement. The objective is therefore to develop a unified framework that forecasts soil strength over a 1–7-day horizon, identifies vehicle-specific GO, SLOW-GO, and NO-GO conditions, and plans routes that minimize travel cost while reducing immobilization risk and unnecessary mission delays.

Each vehicle band b is classified for day $t+h$ from the forecast soil strength $\widehat{\mathrm{RCI}}_{t+h}$, less a safety margin $m_{b,h}$:

$$c_{b,t+h} = \begin{cases} \text{NO-GO}, & \widehat{\mathrm{RCI}}_{t+h} - m_{b,h} < \mathrm{VCI}_1, \\ \text{SLOW-GO}, & \widehat{\mathrm{RCI}}_{t+h} - m_{b,h} < \mathrm{VCI}_{50}, \\ \text{GO}, & \text{otherwise}. \end{cases} \tag{1}$$

where VCI_1 and VCI_{50} are the soil strengths the band needs for one and for 50 passes in one track (Section IV).

IV. Proposed Approach

Fig. 1 shows the framework; RCI and VCI are both in psi, and routes are planned for the day of travel rather than the day of planning.

A. Terrain–Vehicle Interaction Model

1) Soil Strength: With the 0–7 cm and 7–28 cm soil moisture scaled to $s_t^{(1)}, s_t^{(2)} \in [0,1]$ between 20 and 49 vol% (the 11-year range, rounded) and R_t^{3d} the rainfall of the last three days, the soil wetness index (SWI) and the soil strength are expressed as:

$$\mathrm{SWI}_t = 0.35\, s_t^{(1)} + 0.65\, s_t^{(2)}, \tag{2}$$

$$\mathrm{RCI}_t = 300\, e^{-k\,\mathrm{SWI}_t} \left(1 - 0.15 \min\!\left(R_t^{3d}/50\,\mathrm{mm},\ 1\right)\right), \tag{3}$$

where $k = \ln(300/15)$. As in the strength–moisture model used in [2], $\log(\mathrm{RCI})$ falls linearly with moisture. The constants are chosen in this work: the air-dry 300 lies in the 100–500 psi excess strength beyond which NRMM treats fine-grained soil as a hard surface [4], and the saturated 15 follows field readings at immobilization [13]. The deeper layer weighs more as it covers both NRMM critical layers, 0–15 cm for one pass and 15–30 cm for 50 [4]; weights of 0.25/0.75 or 0.5/0.5 change the daily class on at most 2% of vehicle-days. The rain factor adds free surface water, which daily soil moisture misses; its 50 mm is the top of the 25–50 mm rainfall class of [2], and its 15% cap is below their 20–50% reductions because soil moisture already carries most of the rain's effect.

2) Vehicle Cone Index: The Mobility Index (MI) [3], [4] of a tracked vehicle is expressed as:

$$\mathrm{MI} = \left(\frac{\mathrm{CPF}\cdot\mathrm{WF}}{\mathrm{TF}\cdot\mathrm{GF}} + \mathrm{BF} - \mathrm{CF}\right)\mathrm{EF}\cdot\mathrm{TrF}, \tag{4}$$

from the contact pressure, weight, track, grouser, bogie, clearance, engine and transmission factors, and gives the strength needed for one and for 50 passes in one track [5]:

$$\mathrm{VCI}_1 = 7.0 + 0.2\,\mathrm{MI} - 39.2/(\mathrm{MI} + 5.6), \tag{5}$$

$$\mathrm{VCI}_{50} = 19.27 + 0.43\,\mathrm{MI} - 125.79/(\mathrm{MI} + 7.08). \tag{6}$$

Wheeled vehicles use the wheeled forms [13], [14] with 15% tire deflection and automatic transmissions; half-tracks take the worse of the track unit (tracked) and the front axle (wheeled, 25% of the weight). As the data give only mass, power and top speed, track dimensions are interpolated by mass through six published reference vehicles (M22 to Tiger II) and tires follow standard sizes per axle load (HMMWV to heavy 8×8). The implementation reproduces the published M1 example (MI 98.6, VCI_1 26.3, VCI_{50} 60.5) [5] and 24 published 8×8 values to within 1.7 psi [14]. Vehicles fall into five VCI_1 bands set by the fleet mix (A: light tracked, B–D: medium to heavy tracked, E: wheeled), each represented by its median VCI_1, VCI_{50}, power-to-weight ratio, contact pressure and top speed.

3) Passes and Speed: The classes of (1) carry the labels of military terrain analysis [1], NO-GO below VCI_1 follows from its definition [3], and GO at VCI_{50} is our mapping. SLOW-GO is quantified by the number of vehicles n that one track carries, expressed as:

$$n(\mathrm{RCI}) = 50^{(\mathrm{RCI}-\mathrm{VCI}_1)/(\mathrm{VCI}_{50}-\mathrm{VCI}_1)}, \tag{7}$$

which is proposed in this work: only the 1-pass and 50-pass points are published [3], so the requirement rises with $\ln n$ between them. The speed follows the NRMM fine-grained soil relations [4], in which the resistance R/W, traction T/W and slip s depend on $\mathrm{RCI} - \mathrm{VCI}_1$. On a slope θ the vehicle is stuck if its traction cannot overcome the grade and the motion

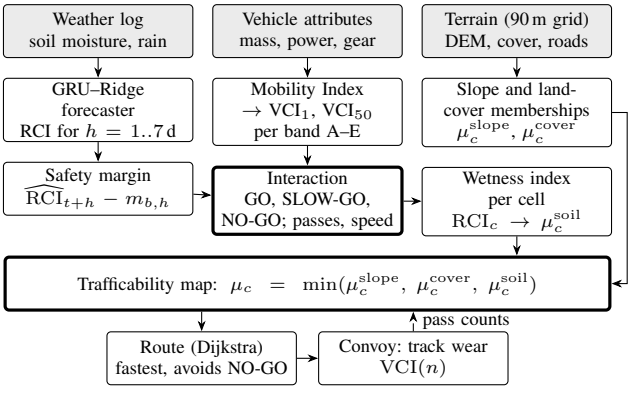


Fig. 1. Proposed framework: the forecaster predicts soil strength from the weather log, the Mobility Index gives each vehicle's cone indices, and the planner combines both with terrain into a per-vehicle map and route.

resistance, $(T/W)\cos\theta < \sin\theta + (R/W)\cos\theta$; otherwise it moves at the speed at which its power, less slip, overcomes them:

$$v = \min\left(\frac{\eta P\,(1-s)}{W\,(\sin\theta + (R/W)\cos\theta)},\ v_{\max} \right), \tag{8}$$

with P the engine power, W the weight and $v_{\max}$ the table's top speed; paved roads use NRMM's hard-surface resistance. Constant power is assumed, as the table has no gearbox data, and $\eta = 0.8$ is an assumed driveline efficiency, to which speed is proportional and NO-GO calls and passes are blind. The NO-GO line stays at VCI_1, 0.7–2.1 psi above NRMM's own level-ground limit. At VCI_1 the relations give the 20% slip at which NRMM defines it, and for the M1 they give 12 km/h at VCI_1 and 45 km/h 20 psi above, comparable to the published NRMM run of about 20–56 km/h [5]. Speeds are upper bounds, as NRMM's vegetation, ride and obstacle limits are not modeled.

B. GRU–Ridge Forecaster

Without numerical weather prediction, the forecaster uses 27 daily channels of the local log (soil moisture, RCI, rainfall, Hargreaves evapotranspiration, meteorology, antecedent precipitation indices with decay 0.85 and 0.95 and seasonal terms). As RCI has a lag-1 autocorrelation of 0.99, it predicts the change for all lead times $h = 1, \ldots, 7$ days at once:

$$\widehat{\mathrm{RCI}}_{t+h} = \mathrm{RCI}_t + \tfrac{1}{2}\left[\bar{g}_h(\mathbf{X}_{t-59:t}) + r_h(\boldsymbol{\phi}_t)\right], \tag{9}$$

where $\bar{g}_h$ averages three GRU networks over a 60-day window, by which the slowest antecedent index has decayed to 5%, and r_h is a ridge regression on 128 causal features (settings in Table II; all recurrent models in Table III share the GRU size). Equation (9) is proposed in this work: predicting the change makes persistence the zero-change forecast, and averaging a nonlinear sequence model with a linear feature model lets their errors partly cancel. Comparing the forecast directly with VCI_1 calls a NO-GO day passable whenever the forecast errs high: on the test years, a tenth of them at 1 day and a quarter at 7 days. The margin $m_{b,h}$ of (1) is a buffer against this error. The error is the same for every vehicle, but it grows with h, and how often it changes a call depends on where the band's threshold lies: through each monsoon the soil hovers around the VCI_1 of light tracked vehicles but stays well below that of wheeled ones (Fig. 2). The margin is therefore set per band and lead time: searched in 0.25 psi steps up to 40 psi, it is the smallest that catches 90% of NO-GO days. This rule, proposed in this work, fixes the miss rate per vehicle class.

C. Trafficability Mapping

The map is built on a 90 m grid of 76,218 cells (3×3 blocks of a 30 m Google Earth Engine export), which holds the mean elevation and slope, the dominant land cover and the peak NDVI over three seasons; cells that are mostly water are impassable. The SWI of each cell is shifted by $\delta_c = 0.1\,\mathrm{clip}(z_c, -2, 2)$, where z_c is the z-score of its topographic wetness index, so that by (3) $\mathrm{RCI}_c = \widehat{\mathrm{RCI}}\,e^{-k\delta_c}$. The scale is chosen in this work: 2.9 vol% per standard deviation, capped at 5.8 vol%. The soil membership is expressed as:

$$\mu_c^{\mathrm{soil}} = \begin{cases} 0, & \mathrm{RCI}_c < \mathrm{VCI}_1, \\ 0.25 + 0.75\min(\rho_c, 1), & \text{otherwise}, \end{cases} \tag{10}$$

where $\rho_c = (\mathrm{RCI}_c - \mathrm{VCI}_1)/(\mathrm{VCI}_{50} - \mathrm{VCI}_1)$. Equation (10) is proposed in this work: the NRMM speed at VCI_1 is 17–25% of that at VCI_{50} for the band vehicles, hence the floor of 0.25, and the linear rise is within 0.1 of the NRMM curve at the midpoint. Slope and land-cover memberships follow the tables of [2], with the slope steps smoothed into a monotone curve. As 79% of the plateau is tree cover, which [2] scores as impassable, forest is mapped by peak NDVI to their forest ($\geq$0.75, about the upper tercile; 0), bush-wood (0.60–0.75; 0 wheeled, 0.2 tracked) and grassland rows ($<$0.60; 0.4 wheeled, 0.6 tracked). Water scores 0, and OpenStreetMap roads from motorway to track score 1 for slope and cover, and paved ones also for soil. Before soil, 26% of the plateau is passable for wheeled and 72% for tracked vehicles. The memberships combine by fuzzy AND, which vetoes any single impassable factor, as the improved map of [2] does:

$$\mu_c = \min\left(\mu_c^{\mathrm{slope}},\ \mu_c^{\mathrm{cover}},\ \mu_c^{\mathrm{soil}}\right). \tag{11}$$

D. Route Planning and Convoys

Each cell is linked to its eight neighbors, NO-GO cells are removed, and an edge costs the time to cross it at a speed proportional to μ:

$$w(u, v) = d_{uv} \cdot \tfrac{1}{2}\left(1/\mu_u + 1/\mu_v\right), \tag{12}$$

where d_{uv} is 90 m, or $90\sqrt{2}$ m on a diagonal; (12) is proposed in this work and charges each half step at the speed of its cell. A trip planned on day t for day $t + h$ forecasts $\widehat{\mathrm{RCI}}_{t+h}$ with (9), subtracts $m_{b,h}$, maps μ_c with (10) and (11), and finds the least-cost route with Dijkstra's algorithm; if none reaches the destination, the trip is held back. For travel-time routing, $1/\mu$ in (12) becomes the time per km at the speed (8), scaled off road by the cover membership. In a convoy each pass weakens the track; inverting (7), the n-th vehicle needs:

$$\mathrm{VCI}(n) = \mathrm{VCI}_1 + (\mathrm{VCI}_{50} - \mathrm{VCI}_1)\ln n / \ln 50. \tag{13}$$

In the wear-aware mode the vehicles plan in turn on the updated pass counts with $\mathrm{VCI}(n)$ in place of VCI_1; in follow-the-leader all take the leader's route. The 3×3-cell staging areas at both ends, where vehicles assemble, are not worn.

V. Experimental Discussions

A. Dataset Description

In this study, the following three datasets are utilized:

1) *Pachmarhi Weather Record:* daily data for the Pachmarhi plateau (Satpura range, 1,067 m), 2015–2025 (4,018 days): soil moisture, temperature and humidity from ERA5-Land [15], wind, pressure and cloud cover from ERA5 [16], and IMD gridded rainfall [17], replaced in

2025 by ERA5, whose day runs 00–24 h rather than 08:30–08:30 IST.

2) *Vehicle Attribute Dataset:* 2,451 records compiled from a simulation game's files [18], of which mass, power and top speed are used. After removing non-vehicles and replacing 111 default masses with public figures, 1,145 ground vehicles remain (1,004 tracked, 125 wheeled, 16 half-tracks); as masses are per-template values, results are reported by band (A: $<$18, B: 18–21, C: 21–24, D: 24–30, E: $\geq$30 psi).
3) *Terrain Data:* a Google Earth Engine [19] export with a 30 m elevation model, ESA WorldCover [20] and seasonal Sentinel-2 [21] NDVI, and 403 OpenStreetMap [22] road segments. Its Sentinel-1 radar layers are not used: under this canopy they mostly measure vegetation.

The record is split into Training (2015–21), Validation (2022–23) and Testing (2024–25), with 2,498, 730 and 724 forecast origins.

B. Evaluation Setup and Metrics

Forecasts use only inputs observed up to each test day, and margins are tuned on one test year and scored on the other. With NO-GO as the positive class of each vehicle-day, recall is fixed at 90% by the margin, so decisions are compared on false NO-GO calls (FP per vehicle-year); forecasts are scored by RMSE and R^2. Table II gives the forecaster's settings and its test error as the training loss: the mean squared error of the RCI change in units of its training variance, on which persistence scores 0.82, 0.77 and 0.82 at 1, 3 and 7 days. For routing, 6 starts and 5 destinations at opposite ends of the largest road network give 30 trips of at least 8 km, planned for every test day, band and lead time $h \in \{1, 3, 7\}$ and scored on the true ground of the travel day: *stuck* if the route crosses a NO-GO cell, *held* if no route is planned although one existed, and the extra time over the best route. The planners are terrain only, today's map (planning on current conditions as [2] does, with our memberships), today's map with a margin, and the forecast with its margin; variants replace the fuzzy AND by a product or by the equal-weight mean of [2] with and without its veto, and a sweep scales both margins from 0 to 3 times. Convoys of ten band D and band E vehicles run on soft days with RCI between $1.05\,\mathrm{VCI}_1$ and $1.5\,\mathrm{VCI}_{50}$ (up to 30 per band, 3 trips each). 95% confidence intervals (CI) come from a moving-block bootstrap over test days (14-day blocks, 2,000 resamples).

TABLE II
HYPERPARAMETERS OF THE GRU–RIDGE FORECASTER (531,228 PARAMETERS IN ALL) AND ITS TEST ERROR, AS THE TRAINING LOSS AT 1/3/7 DAYS. THE RIDGE MODELS ARE REFIT ON 2015–23 ONCE α IS CHOSEN.

Hyperparameter	**GRU (3 seeds)**	**Ridge**
No. of neurons	128 per layer; head 128, 7 outputs	None (linear on 128 features), 7 outputs
No. of layers	2 GRU (dropout 0.2) + 2 dense	1 linear per lead time
Loss function	MSE on ΔRCI, standardized per lead time	Squared error on ΔRCI with an L^2 penalty α
Optimizer	AdamW, LR 0.001 (one-cycle), weight decay 0.0001, gradient clip 1.0	Closed form; α from 10^{-1} to 10^{4} per lead time, chosen on validation
Batch size	128	Full data
Activation function	Sigmoid and tanh (gates), ReLU (head)	None (linear)
Epochs	300, early stopping (patience 30)	–
Train–test splitting	Train 2015–21, validation 2022–23, test 2024–25	

L^2 error (training loss, test)		1 day	3 days	7 days
	GRU	0.688	0.657	0.580
	Ridge	0.727	0.645	0.584
	GRU–Ridge (9)	**0.668**	**0.593**	**0.523**

TABLE III
FORECASTING MODELS TESTED, 2024–25: RMSE (PSI) BY LEAD TIME AND FALSE NO-GO CALLS PER VEHICLE-YEAR (FP/YR) AT 90% RECALL.

		RMSE			FP/yr	
Model	Params	1 d	3 d	7 d	3 d	7 d
Persistence	–	6.71	15.16	26.02	30.4	45.8
Climatology	–	35.60	35.47	34.73	23.4	**23.4**
ARIMAX	14	**5.69**	13.88	22.16	22.9	29.5
Ridge	0.9k	6.31	13.86	21.92	23.2	31.2
Random Forest	–	6.55	14.93	23.89	25.4	32.7
LightGBM	–	6.33	14.51	24.84	25.6	32.5
LSTM [11]	690k	7.05	15.46	22.58	22.8	35.7
GRU	530k	6.14	13.98	21.85	24.8	35.5
TCN	370k	5.84	13.66	22.89	23.1	35.7
Transformer [12]	1.85M	6.09	13.88	22.61	**19.7**	34.6
TFT	3.47M	6.05	**12.83**	21.18	23.2	35.7
N-BEATS	2.29M	6.15	13.50	**20.68**	24.1	32.4
Ensemble (top 3)	2.75M	5.82	13.34	21.41	21.6	34.4
GRU–Ridge (ours)	531k	6.05	13.29	20.76	24.0	31.4

C. Experimental Results

1) Model Selection: Twenty-two forecasting models and rules, ours included, were tested on the same inputs and years. Table III lists fourteen, with parameters counted over all three seeds; the other eight, all with a higher 7-day RMSE than GRU–Ridge, are seasonal naive, ARIMA, ExtraTrees, LightGBM with linear leaves, a bidirectional GRU with attention (point and quantile versions), and the 10th-percentile (P10) rules of the TFT and of that GRU. The GRU–Ridge forecaster has the second-lowest RMSE averaged over the seven lead times (14.64 psi, after 14.45 for the TFT) and the fewest false NO-GO calls of the deep models at 7 days, with a sixth of the TFT's parameters; each member alone is worse at every lead time. Climatology with a margin makes the fewest false calls at 7 days, as NO-GO days a week ahead are largely seasonal.

2) Results by Vehicle Band: Fig. 2 shows why decisions are made per band: the same soil strength crosses each band's VCI_1 at a different time. Each monsoon it drops by over 100 psi within two weeks to near the saturated strength and recovers over about two months, so that bands with a higher VCI_1 return to GO later; wheeled vehicles (band E) are NO-

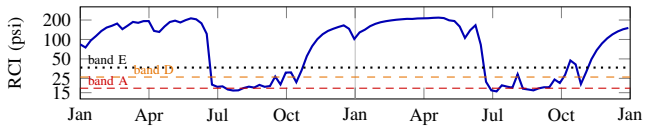


Fig. 2. Weekly mean soil strength over the 2024–25 test years against the one-pass cone index VCI_1 of bands A (light tracked), D (heavy tracked) and E (wheeled). Below a line, that band is NO-GO.

TABLE IV
RESULTS BY VEHICLE BAND, 2024–25: SHARE OF DAYS NO-GO AND SLOW-GO, MEDIAN PASSES AND LEVEL-GROUND SPEED ON SLOW-GO DAYS, AND FALSE NO-GO CALLS PER VEHICLE-YEAR AT 7 DAYS FOR PERSISTENCE (P) AND THE FORECASTER (F).

				SLOW-GO days		FP/yr	
Band	VCI_1	NO-GO	SLOW-GO	Passes	km/h	P	F
A (tracked)	17.6	13.3%	21.8%	1.8	14.2	74.0	**54.3**
B (tracked)	19.0	18.9%	16.8%	2.4	13.9	54.0	**34.0**
C (tracked)	21.6	24.4%	12.9%	2.7	13.8	43.9	**27.8**
D (tracked)	26.2	29.0%	9.8%	3.4	18.6	28.6	**22.0**
E (wheeled)	36.7	33.9%	8.6%	5.2	31.7	13.8	**7.6**

GO on 34% of days and light tracked ones (band A) on 13% (Table IV). On a typical SLOW-GO day only 2–5 vehicles can use one track, at 14–32 km/h against 57–60 km/h on dry ground (Table IV); at 5 psi above VCI_1, a track carries 1.5–2.3 vehicles at 12–16 km/h. At equal recall, the forecast makes 23–45% fewer false NO-GO calls than the margin rule on persistence in every band, and pooled over all vehicles 21% fewer at 3 days (CI 4–37%) and 31% at 7 days (CI 14–46%). Light tracked vehicles (band A) are the hardest case, as their VCI_1 is close to the saturated strength (Fig. 2).

3) Route Planning: Over 325,800 trips per planner, planning on terrain alone sends 35.5% of trips into NO-GO ground. At 7 days, the forecast beats today's map on both errors: 2.9% against 4.0% stuck (27% fewer, CI 11–54%) and 4.5% against 5.8% held (CI 6–37% fewer). At 3 days it gives 1.6% against 2.1% stuck and 2.3% against 3.0% held, but both intervals include zero. A margin on today's map is slightly safer at its tuned setting but holds back far more trips; the forecast holds back 47% fewer at both lead times (CI 25–62% at 3 days, 27–61% at 7 days). Fig. 3 traces this trade-off as each margin is scaled: at both lead times the forecast's curve lies below and to the left of that of today's map, and at an equal stuck rate it holds back 40% fewer trips at 3 days (CI 3–67%) and 31% at 7 days, where the interval (−2–69%) includes zero. Completed routes are within 0.1% of the best travel time. The product rule matches the fuzzy AND, the equal-weight mean with veto sends 1.8% into NO-GO ground at 3 days but takes 10.4% longer, and without the veto 34.8%. Routing on travel time shortens trips by about 5% at the same stuck rate (30.8 against 32.4 minutes at 3 days), with predicted times on average within 0.3% (3 days) and 1.0% (7 days) of the travel day's, and 9 in 10 within −2.1% to +3.4%. With no weather forecast as input, no planner foresees the day the monsoon breaks; the gain comes around the transitions.

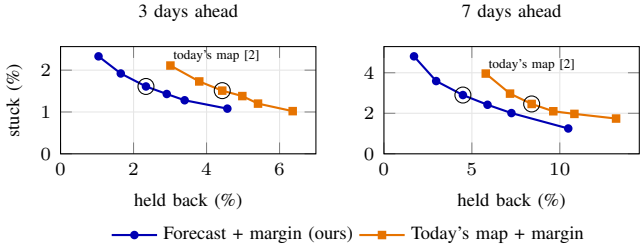


Fig. 3. Routing backtest, 2024–25: trips sent into NO-GO ground (stuck) against possible trips held back, as each planner's safety margin is scaled from zero (today's map, no margin) to three times its tuned value (ringed). Lower left is better.

TABLE V
COMPARISON WITH PAST WORKS. SM: SOIL MOISTURE; CI: CONE INDEX; TM: TRAFFICABILITY MAP. EACH R^2 IS ON THE WORK'S OWN TARGET.

					R^2		
Work	Model	Target	Vehicle	Route	1 d	3 d	7 d
[2]	AHP map	TM	✓	✓	–	–	–
[11]	RF	SM	×	×	0.929	0.891	0.832
[11]	LSTM	SM	×	×	0.963	0.916	0.842
[11]	Transformer	SM	×	×	0.952	0.912	0.843
[11]	GAN-LSTM	SM	×	×	0.963	0.919	0.846
[12]	Transformer	SM	×	×	0.523 (mean)		
Ours	**GRU–Ridge**	RCI	✓	✓	**0.993**	**0.968**	**0.922**

4) Convoy Coordination: A convoy of ten heavy tracked vehicles (band D) following its leader completes the route with 5.5 vehicles on average, as the leader's track fails under the vehicles behind; the wear-aware mode completes it with 9.2. Wheeled vehicles (band E) gain less (6.8 to 7.3), as the forest leaves them few parallel lanes.

5) Comparison with Past Works: Table V compares the framework with published results; each R^2 is on the work's own target and data. The forecaster reaches R^2 of 0.993, 0.968 and 0.922 on soil strength at 1, 3 and 7 days, higher at every lead time than all soil-moisture models of [11], including their best, the GAN-LSTM, and than the Transformer of [12]. It is the only approach that combines a multi-day forecast with per-vehicle assessment and routing, and the route-planning results above show its gain over the current-condition planning of [2].

D. Limitations

Soil strength is computed from modeled soil moisture for one 9 km cell, not measured, and its spread by wetness index is unchecked in the field. Vehicle masses come from game files and the running gear is assumed, so results are reported by band. The pass and wear curves and the speed assumptions are ours, so speeds are upper bounds. Margins tuned on the two test years differ by up to 7.25 psi, and two test years leave some intervals wide.

VI. Conclusion

This study presents an integrated framework for forecasting terrain trafficability and planning safe, efficient routes for heterogeneous military ground vehicles under changing environmental conditions. The proposed framework combines a hybrid Gated Recurrent Unit–Ridge (GRU–Ridge) model for 1–7-day soil strength forecasting with vehicle-specific cone index requirements, terrain characteristics, fuzzy trafficability mapping, and Dijkstra-based route optimization. Convoy coordination is further incorporated to account for the progressive degradation of soil bearing capacity caused by repeated vehicle passage.

Experiments using meteorological, vehicle, and geospatial data for the Pachmarhi plateau demonstrate the potential of the proposed approach. The GRU–Ridge model, with a sixth of the parameters of the only tested model with a lower average error, achieves an average RMSE of 14.64 psi across the seven forecast horizons and an R^2 of 0.922 for soil strength at a seven-day horizon. Compared with route planning based on current terrain conditions, the forecast-based approach reduces the proportion of trips encountering impassable terrain by 27% and of trips held back unnecessarily by 22% at seven days. Furthermore, incorporating track degradation into convoy planning increases the average number of heavy tracked vehicles that complete a route from 5.5 to 9.2 out of 10. These findings highlight the benefits of integrating predictive soil strength assessment with vehicle-specific mobility constraints and route optimization.

Future work will focus on validating the predictions using field cone-index measurements, improving vehicle parameter estimation, incorporating numerical weather forecasts, and evaluating the framework across additional terrains and seasonal conditions. Overall, the proposed approach provides a promising basis for anticipatory, risk-aware military mobility planning under dynamic environmental conditions.

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Data Availability

Data will be available on request.

References

[1] M. N. Khan, M. Kashif, and A. Shah, "Off-road trafficability for military operations using multi-criteria decision analysis," International Journal of Advanced Remote Sensing and GIS, vol. 10, no. 1, pp. 3425–3437, 2021.

[2] Q. Liu, X. You, X. Zhang, and J. Zuo, "Construction of dynamic trafficability map for unmanned vehicles considering multiple environmental factors and path planning," Scientific Reports, vol. 15, p. 9957, 2025.

[3] A. A. Rula and C. J. Nuttall, Jr., "An analysis of ground mobility models (ANAMOB)," U.S. Army Engineer Waterways Experiment Station, Vicksburg, MS, Tech. Rep. M-71-4, 1971.

[4] P. W. Haley, M. P. Jurkat, and P. M. Brady, Jr., "NATO reference mobility model, Edition I, users guide, volume I," U.S. Army Tank-Automotive Research and Development Command, Warren, MI, Tech. Rep. 12503, 1979.

[5] T. Ciobotaru, "Semi-empiric algorithm for assessment of the vehicle mobility," Leonardo Electronic Journal of Practices and Technologies, vol. 8, no. 15, pp. 19–30, 2009.

[6] A. Dutta, M. Banerjee, and R. Ray, "Land capability assessment of Sali watershed for agricultural suitability using a multi-criteria-based decision-making approach," Environmental Monitoring and Assessment, vol. 196, no. 3, p. 237, 2024.

[7] O. Mema, E. Jojie, P. Laze, V. Vorpsi, N. Vata, and I. Salillari, "Optimizing land management: A GIS-based model for sustainable agricultural machinery operation in Frakulla administrative unit, Albania," International Journal of Geoinformatics, vol. 22, no. 3, pp. 51–69, 2026.

[8] H. Gao, W. Huang, and X. Yang, "Applying probabilistic model checking to path planning in an intelligent transportation system using mobility trajectories and their statistical data," Intelligent Automation & Soft Computing, vol. 25, no. 3, 2019.

[9] Z. Xu et al., "Multi-objective optimization for multimodal multi-objective multi-point shortest path problem considering unforeseeable road eventualities," IEEE Transactions on Intelligent Transportation Systems, 2025.

[10] P. Gupta, B. Pareek, G. Singal, and D. V. Rao, "Edge device based military vehicle detection and classification from UAV," Multimedia Tools and Applications, vol. 81, no. 14, pp. 19813–19834, 2022.

[11] Y. Wang, L. Shi, Y. Hu, X. Hu, W. Song, and L. Wang, "A comprehensive study of deep learning for soil moisture prediction," Hydrology and Earth System Sciences, vol. 28, pp. 917–943, 2024.

[12] Y. Wang and Y. Zha, "Comparison of transformer, LSTM and coupled algorithms for soil moisture prediction in shallow-groundwater-level areas with interpretability analysis," Agricultural Water Management, vol. 305, p. 109120, 2024.

[13] P. Roşca, T. Ciobotaru, G. Olaru, and C. Puică, "Considerations regarding vehicle cone index for fine-grained soil," Journal of Military Technology, vol. 1, no. 2, pp. 39–44, 2018.

[14] P. Roşca and T. Ciobotaru, "8×8 armoured personnel carrier mobility evaluations using vehicle cone index," MTA Review, vol. 27, no. 2, pp. 77–82, 2017.

[15] J. Muñoz-Sabater et al., "ERA5-Land: A state-of-the-art global reanalysis dataset for land applications," Earth System Science Data, vol. 13, pp. 4349–4383, 2021.

[16] H. Hersbach et al., "The ERA5 global reanalysis," Quarterly Journal of the Royal Meteorological Society, vol. 146, no. 730, pp. 1999–2049, 2020.

[17] D. S. Pai, L. Sridhar, M. Rajeevan, O. P. Sreejith, N. S. Satbhai, and B. Mukhopadhyay, "Development of a new high spatial resolution ($0.25^\circ \times 0.25^\circ$) long period (1901–2010) daily gridded rainfall data set over India and its comparison with existing data sets over the region," Mausam, vol. 65, no. 1, pp. 1–18, 2014.

[18] Gaijin Entertainment, "War Thunder," video game, 2016.

[19] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau, and R. Moore, "Google Earth Engine: Planetary-scale geospatial analysis for everyone," Remote Sensing of Environment, vol. 202, pp. 18–27, 2017.

[20] D. Zanaga et al., "ESA WorldCover 10 m 2021 v200," Zenodo, 2022, doi: 10.5281/zenodo.7254221.

[21] M. Drusch et al., "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," Remote Sensing of Environment, vol. 120, pp. 25–36, 2012.

[22] M. Haklay and P. Weber, "OpenStreetMap: User-generated street maps," IEEE Pervasive Computing, vol. 7, no. 4, pp. 12–18, 2008.